Lecture 23

Discrete series and the rank one unitary picture

Objectives. *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules. Finally we state the full list of irreducible unitary representations.*

By the end of this lecture you will be able to:

- Prove that Π_n is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its K-types $n, n+2, \dots$ and the action of H', E, F.
- Decide which lowest weights λ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification and derive the positivity bounds behind it.

23.1 The disk model

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation Π, a vector v has *K-type k* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all θ. If such a v is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for a, b, c, d shows that φ is an isomorphism of Lie groups from G onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by γ is complex linear on all 2×2 matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}, \qquad j(h, z) = \bar\beta z + \bar\alpha.$$

Lemma 23.1. *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

(a) $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

(b) h *maps* D *into* D *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

(c) $d(h \cdot z)/dz = j(h,z)^{-2}$.

Proof. (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so h maps D into D. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So h maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar\beta_1(\alpha_2 z + \beta_2) + \bar\alpha_1(\bar\beta_2 z + \bar\alpha_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z, 1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z, 1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar\beta z + \bar\alpha) - \bar\beta(\alpha z + \beta) = 1$. ∎

Definition 23.2. Let $n \geq 2$ be an integer and $dA = dx\,dy$. For f holomorphic on D put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1-|z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic f on D with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in D with smooth boundary.

Lemma 23.3. *If $f(z) = \sum_{m \geq 0} a_m z^m$ is holomorphic on D, then*

$$\|f\|_n^2 = \sum_{m \geq 0} c_m |a_m|^2, \qquad c_m = \|z^m\|_n^2 = \frac{m!\,(n-1)!}{(n+m-1)!} = \binom{n+m-1}{m}^{-1}.$$

So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m = c_m^{-1/2} z^m$. Moreover $\|g\|_n \leq \sup_D |g|$ for bounded holomorphic g.

Proof. On $|z| = s < 1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi} |f(se^{i\theta})|^2 d\theta = 2\pi \sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi} \int_{|z| \leq r} |f|^2 (1-|z|^2)^{n-2} dA = \sum_m |a_m|^2\, (n-1) \int_0^r 2s^{2m+1} (1-s^2)^{n-2} ds.$$

The terms are nonnegative and increase with r, so we may let $r \to 1$ termwise. With $t = s^2$ and repeated integration by parts,

$$(n-1) \int_0^1 t^m (1-t)^{n-2} dt = (n-1) \frac{m!\,(n-2)!}{(n+m-1)!} = c_m.$$

Conversely, let $(a_m)_{m \geq 0}$ be complex numbers with $C = \sum_m c_m |a_m|^2 < \infty$. Then $|a_m|^2 \leq C \binom{n+m-1}{m} \leq C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on D. By the power-series import, its sum f is holomorphic on D. The formula above then gives $\|f\|_n^2 = C$. Hence $f \mapsto (c_m^{1/2} a_m)_m$ is an isometry of $\mathcal{A}_n$ onto ℓ^2. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2 \leq \sup |g|^2\, \|1\|_n^2 = \sup |g|^2$, since $c_0 = 1$. ∎

Theorem 23.4. *For $h \in \mathrm{SU}(1,1)$ and $f \in \mathcal{A}_n$ define*

$$(\Pi_n(h) f)(z) = j(h^{-1}, z)^{-n}\, f(h^{-1} \cdot z). \tag{23.5}$$

Then $g \mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of G on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.

Proof. Homomorphism. Apply the cocycle identity of Lemma 23.1(b) to h_2^{-1} and h_1^{-1}:

$$(\Pi_n(h_1)\Pi_n(h_2) f)(z) = j(h_1^{-1}, z)^{-n}\, j(h_2^{-1}, h_1^{-1} \cdot z)^{-n} f(h_2^{-1} h_1^{-1} \cdot z) = (\Pi_n(h_1 h_2) f)(z).$$

Isometry. Substitute $z = h\cdot w$. The cocycle identity with $h^{-1}h = I$ gives $j(h^{-1}, h\cdot w) = j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\ &\quad= |j(h,w)|^{2n}|f(w)|^2\,\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\,\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R \subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over R.

The map $w \mapsto h\cdot w$ is a bijection of D with inverse $w \mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n = \|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

Continuity. Let f, g be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n \geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood U of I. So $h \mapsto \langle g, \Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\| = 1$, so an $\varepsilon/3$ argument extends this to all f, g. Then

$$\|\Pi_n(h)f - f\|_n^2 = 2\|f\|_n^2 - 2\,\mathrm{Re}\langle f, \Pi_n(h)f\rangle_n \to 0 \qquad (h \to I).$$

Unitarity moves this to every other point. ∎

Remark 23.5. Put $\delta = \mathrm{diag}(1,-1)$. Since $\bar\gamma = \gamma\delta$, the matrix $\bar\gamma$ in place of γ gives $\Pi_n^-(g) = \Pi_n(\delta g\delta^{-1})$. Conjugation by δ preserves G, although $\delta \notin G$. Since $\delta k_\theta \delta^{-1} = k_{-\theta}$, the K-types of Π_n^- are the negatives of those of Π_n. Which one is called holomorphic depends on the Cayley matrix and the sign of the K-angle. We label them by K-types.

23.2 K-types and Lie algebra operators

Proposition 23.6. *For all θ and $m \geq 0$,*

$$\Pi_n(k_\theta)z^m = e^{i(n+2m)\theta}z^m.$$

Hence the K-types of Π_n are $n, n+2, n+4, \dots$, each with multiplicity one, and the K-finite vectors are the polynomials.

Proof. By the formula (23.2), the element $h = \varphi(k_\theta)$ has $h^{-1} = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. So $h^{-1}\cdot z = e^{2i\theta}z$ and $j(h^{-1}, z) = e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer k define

$$P_k f = \frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m = e_m$ if $k = n + 2m$, and $P_k e_m = 0$ otherwise. Being bounded, P_k is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of K-type k.

If v is K-finite, its orbit spans a finite-dimensional space U. Every $P_k v$ lies in U, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of v is therefore a polynomial. Conversely, every polynomial is a finite sum of K-types. ∎

Lemma 23.7. *Polynomials are smooth vectors for Π_n. For $Y = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathfrak{su}(1,1)$ and a polynomial f,*

$$d\Pi_n(Y) f = n(rz + s) f + \bigl(rz^2 + (s - p)z - q\bigr) f'. \tag{23.6}$$

Proof. By the lower bound for $|j(h, z)|$ in Lemma 23.1(a), $(h, z) \mapsto (\Pi_n(h)f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in h of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So f is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y)f$ is the pointwise t-derivative at 0 of $(\Pi_n(h_t)f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz + s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1 - tp)z - tq}{1 - t(rz + s)} = z + t\bigl(rz^2 + (s - p)z - q\bigr) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz + s) + O(t^2)$, and the product and chain rules give (23.6). ∎

Proposition 23.8. *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H') z^m &= (n + 2m) z^m, \\ d\Pi_n(E) z^m &= (n + m) z^{m+1}, \\ d\Pi_n(F) z^m &= -m\, z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F) 1 = 0. \end{aligned}$$

The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n - 2)$.

Proof. The right side of (23.6) is complex linear in (p, q, r, s). So it also computes $d\Pi_n$ on the matrices in (23.3). For H' we have $p = -1$, $s = 1$, giving $nf + 2zf'$. For E we have $r = 1$, giving $nzf + z^2 f'$. For F we have $q = 1$, giving $-f'$. Apply these to z^m. Since $[E, F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On z^m its eigenvalue is

$$(n + 2m)^2 + 2(n + 2m) - 4(n + m)(m + 1) = n^2 - 2n.$$ ∎

For a unitary representation Π and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices H_0, S and W are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators H_0 and S. It also turns the coefficient i of S into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors u and v satisfy

$$\langle d\Pi(E)u, v\rangle = -\langle u, d\Pi(F)v\rangle, \qquad \langle d\Pi(H')u, v\rangle = \langle u, d\Pi(H')v\rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for c_m in Lemma 23.3. Conversely, distinct K-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

Theorem 23.9. *Π_n is irreducible for every integer $n \geq 2$.*

Proof. Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each P_k maps V into V, since its Riemann sums do and V is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some e_m lies in V. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in V. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in V. By the operator formulas of Proposition 23.8, E sends z^m to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, F sends z^m to $-mz^{m-1}$ with $m \neq 0$. So every z^k lies in V. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on G. The family Π_n, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of e_0.

23.3 Descent conditions

Proposition 23.10. *Let Π be a strongly continuous representation of G on a Hilbert space. Let $v \neq 0$ have K-type k.*

(a) *$k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such Π has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its K-finite part.*

(b) *$\Pi(-I)v = (-1)^k v$.*

(c) *$\Pi_n(-I) = (-1)^n \mathrm{id}$. Hence Π_n factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if n is even.*

Proof. (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight K-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of K-type λ, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of K-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I, z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So Π_n factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every K-type $n+2m$ of Π_n has the parity of n. The integrality argument of Proposition 23.10(a) uses the period 2π of K in G. On the universal covering group the lift of K is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

Remark 23.11. The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure μ on D gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on D. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

23.4 Comparison and the unitary picture

Let V be a $(\mathfrak{g}, K)$-module of G (Lecture 21) on which Ω acts by a scalar c. Write v_k for a vector of K-type k. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

Example 23.12. Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In Π_n the K-types are $n, n+2, \ldots$. F kills z^0, and E is injective.

(ii) Put $d = n - 2$ and replace n by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends f to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most d. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills z^d. The K-types are $-d, \ldots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a K-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n - 1$, and with ε chosen so that its K-types have the parity of n. By Lecture 22, Exercise 22.4, Ω acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$. So the module is reducible. It has a filtration whose subquotients have K-types $k \leq -n$, $2 - n \leq k \leq n - 2$ and $k \geq n$. These subquotients have the K-types and Casimir value of Π_n^-, of the finite-dimensional module Π_{-d} with $d = n - 2$, and of Π_n.

Proposition 23.13. *Let V be an irreducible $(\mathfrak{g}, K)$-module of G. Assume each K-type has multiplicity at most one and Ω acts by c. Suppose V has a positive definite inner product satisfying (23.7) for the module action. If k and $k+2$ are K-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

Consequently:

(i) if the K-types are $\lambda, \lambda+2, \lambda+4, \ldots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;

(ii) if the K-types are $2\mathbb{Z}$, then $c < 0$;

(iii) if the K-types are $2\mathbb{Z}+1$, then $c < -1$.

Proof. By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that c is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the K-type spaces of that parity is a submodule. By irreducibility, all K-types have

the parity of k. Since E raises and F lowers K-types by 2, the span of the v_j with $j \le k$ is invariant. It is nonzero and misses v_{k+2}, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda - 2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda} v_\lambda$, which makes λ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even k and -1 over odd k. ∎

Twisting by δ from Remark 23.5 interchanges E, F, sends H' to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \ge 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \ldots$.

Theorem 23.14 (Bargmann; cited). *Every irreducible unitary representation of G is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its K-types and its Casimir value c. Here Π_1 means formula (23.5) at $n = 1$ on the completion of polynomials in the norm $\sum |a_m|^2$. The unitarity and irreducibility of Π_1 are part of this cited theorem.*

Family	*K-types*	c
trivial	$\{0\}$	0
even principal series	$2\mathbb{Z}$	$c \le -1$
odd principal series	$2\mathbb{Z} + 1$	$c < -1$
complementary series	$2\mathbb{Z}$	$-1 < c < 0$
Π_n, $n \ge 1$	$n, n+2, \ldots$	$n(n-2)$
Π_n^-, $n \ge 1$	$-n, -n-2, \ldots$	$n(n-2)$

For $n \ge 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.

Proposition 23.13 proves the necessity of the c-ranges for these K-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its K-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z} + 1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the L^2 norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its K-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one K-types on which Ω is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of G. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

Since G is simple and noncompact, its only finite-dimensional unitary irreducible representation is trivial. Exercise 23.3 checks this directly for the finite-dimensional modules Π_{-d} of Example 23.12(ii).

23.5 Self-checks

1. For $n = 3$, compute c_1 and c_2. Check (23.7) for $u = z$ and $v = z^2$.

   Answer. $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does Π_5 factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   Answer. No, since $\Pi_5(-I) = -\mathrm{id}$. By the K-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with K-types $1, 3, 5, \dots$ and $c = -\frac{1}{2}$?

   Answer. No. By the cited Harish-Chandra facts after Theorem 23.14, the K-finite vectors of such a representation satisfy the hypotheses of Proposition 23.13. Its lowest-weight case (i) shows that the lowest K-type 1 forces $c = 1 \cdot (1-2) = -1$.

23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation Π_n on $\mathcal{A}_n$. Its K-types are $n, n+2, \dots$, each once. The operator E raises the K-type by 2 and F lowers it by 2. The Casimir element Ω acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral K-types. The representation Π_n descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even n. In an irreducible unitary module with K-types $\lambda, \lambda+2, \dots$, the lowest weight is an integer $\lambda \geq 1$, and Ω acts by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.